

## SOFTWARE

### Programming & HPC

- C/C++
- CUDA C++

### Data Analysis

- Python
- Basics of PostgreSQL

### Simulation / Design

- Autodesk Fusion
- COMSOL

### Development Tools

- Linux OS
- Git
- Claude Code

### Typesetting

- L<sup>A</sup>T<sub>E</sub>X

## LANGUAGES

- Polish – Native
- English – Fluent
- German – Basic

## CERTIFICATES

- ACERT Business English C1

## INTERESTS

### Scientific

- Digital modeling of reality

### Non-scientific

- Climbing
- Hiking
- 3D printing and 3D graphics

## PROFILE

Undergraduate student in Nanoengineering of Materials with a strong interest in computational physics, numerical simulations, and theoretical aspects of materials science. Experienced in C/C++, Python and CUDA for scientific computing. Seeking to develop skills in simulation, modeling and data analysis.

## EDUCATION



### Student

Cracow, Poland

AGH University of Krakow

Oct. 2023 – ongoing

Faculty of Physics and Applied Computer Science  
Faculty of Materials Science and Ceramics  
Field: Nanoengineering of Materials  
Specialization: Modeling

## JOB EXPERIENCE

### Data Analyst

at The Henryk Niewodniczański Institute of Nuclear Physics

Cracow, Poland Jul. 2026 – Sep. 2026 (3 months)  
Analysed decades of neutron-monitor and earthquake catalogue data in Python, applying Monte Carlo methods to test for correlations between cosmic-ray flux and seismic activity.

### Production line worker

at YK Poland Sp. z o.o.

Modlniczka, Poland Jul. 2024 – Sep. 2024 (3 months)  
Executed the complete production cycle of metal photo frames, including profile cutting, assembly, shrink-wrapping, and final packaging.

### Production line worker

at BMK Professional Electronics GmbH

Augsburg, Germany Jul. 2023 – Sep. 2023 (3 months)  
Performed manual PCB assembly, operated the reflow soldering process, and handled board preparation involving masking and machine-assisted dispensing.

## ACADEMIC PROJECTS

### Mass Transport Simulation

Developed a GPU-accelerated Computational Fluid Dynamics (CFD) engine to simulate incompressible flow and mass transport using the **Navier-Stokes** and **Advection-Diffusion** equations. Implemented the Stream Function-Vorticity formulation using Finite Difference Methods, solving the resulting linear systems via **parallelized Red-Black Gauss-Seidel relaxation** custom **CUDA** kernels. The simulation features a **Crank-Nicolson scheme** for stable time-stepping of mass distribution and integrates a **Python/FFmpeg** pipeline to render high-framerate visualizations of velocity fields and concentration heatmaps.

### (2+1)D Wave Equation Simulation

Developed a high-performance visualizer for the (2+1)D wave equation by deriving an exact analytical solution via **separation of variables** and **Fourier series expansion**. Implemented the numerical evaluation of the double infinite series using **CUDA C++**, optimizing performance by parallelizing the summation loops on the GPU to handle high-resolution spatial grids. The project features a fully automated pipeline that orchestrates the C++ backend, **Python (Matplotlib)** for 3D surface plotting, and **FFmpeg** for video rendering using a custom **Makefile**.